

OPERATION FALSE ASSURANCE

Team Name	BYEBUST
Phase Number / Title	Phase 4: Interactive Control and Long-Lived External Sessions (P4-06)
Incident Date	2025-10-08
Date Submitted	2026-08-22

No.	Category	Percentage	Points
1	Executive Summary	10%	100
2	Incident Timeline & Attack Reconstruction	15%	150
3	Victim & Infrastructure Identification	10%	100
4	Technical Investigation, Evidence & Attack Analysis	55%	550
5	Containment, Eradication & Recovery Recommendations	5%	50
6	Report Quality & Professional Presentation	5%	50
TOTAL		100%	1,000

1. EXECUTIVE SUMMARY

During Phase 4 of Operation False Assurance (Interactive Control and Long-Lived External Sessions), forensic network session aggregation was performed within Security Onion Hunt to quantify the exact volume of distinct outbound TCP connection streams established to external command-and-control server `86.159.48.25:2222`. Utilizing the hunting aggregation query `destination.ip:"86.159.48.25" AND destination.port:2222 | groupby event.dataset network.transport`, the forensic analysis confirmed exactly **41 distinct TCP connection sessions** recorded in the `zeek.conn` dataset. Each individual connection was mirrored by a corresponding Suricata NIDS alert (41 total alerts) for Dridex JA3 hash beaconing, demonstrating a consistent, automated heartbeat cadence across the interactive C2 lifecycle.

2. INCIDENT TIMELINE AND ATTACK RECONSTRUCTION

Date / Time (UTC)	Source	Destination	Protocol / Port	Activity / Observation
2025-10-08 09:27:17	10.11.28.101	86.159.48.25:2222	TCP / TLS	Initial connection session (#1 of 41) established to external C2 node.
2025-10-08 18:18:00	10.11.28.101:49974	86.159.48.25:2222	TCP / TLS	Mid-phase beacon session (#33 of 41); connection state <code>SF</code> logged in Zeek.
2025-10-08 18:28:35	10.11.28.101:50009	86.159.48.25:2222	TCP / TLS	Automated beacon cycle (#36 of 41); 3-minute interval maintained.
2025-10-08 18:38:11	10.11.28.101:50124	86.159.48.25:2222	TCP / TLS	Pre-final beacon session (#40 of 41); flow UID <code>CH1fzQ3KFWT1UnGU9</code> .
2025-10-08 18:41:20	10.11.28.101:50127	86.159.48.25:2222	TCP / TLS	Final connection session (#41 of 41); flow UID <code>CvghJ14KzEayP8ZNWg</code> .

Attack Reconstruction

Throughout the duration of the compromised session, the malware implanted on workstation `10.11.28.101` repeatedly established distinct TCP connections to `86.159.48.25:2222`. Analysis of the Zeek connection dataset (`zeek.conn`) revealed that each interaction opened a discrete, ephemeral local socket (ports 49974 through 50127) to initiate a full TCP handshake, negotiate a TLSv1.2 encrypted tunnel, exchange telemetry/commands, and gracefully terminate with a connection state of `SF` (SYN/FIN normal close). The cumulative volume of these distinct TCP connection handshakes totaled exactly **41 sessions**,

producing an identical tally of 41 Suricata NIDS alerts and 41 Zeek SSL handshakes (124 aggregate events across all index types including 1 notice and 1 SSL event).

3. VICTIM AND INFRASTRUCTURE IDENTIFICATION

Affected Host

IP Address	Hostname	User Account	Operating System	AD Domain	System Role	Status
10.11.28.101	DESKTOP-CLIENT1	Not Determined	Windows Workstation	mytallbeer.com	Client Endpoint	Active C2 Source (41 Sessions)

Threat Infrastructure

Type	Domain / IP Address	Port	Purpose / Observation
External C2 Server	86.159.48.25	2222 (TCP)	Adversary C2 node receiving 41 distinct TCP connection sessions.
Compromised Endpoint	10.11.28.101	Dynamic (49974–50127)	Internal victim host originating all 41 TCP/TLS connection streams.
Internal Domain Controller	10.11.28.2	53, 88, 139, 389, 445	Active Directory DC (`WIN-FGG5B8C1BYF`) targeted by client during reconnaissance.

4. TECHNICAL INVESTIGATION, EVIDENCE AND ATTACK ANALYSIS

4.1 Analyst Findings

No.	Finding / Objective	Answer / Result	Evidence Ref.
1	Distinct TCP Connection Count (P4-06)	CTK{41} (41 distinct connections)	E-01, E-02
2	Target C2 IP & Port	86.159.48.25:2222	E-01, E-03
3	Primary Connection Log Source	zeek.conn (Count: 41)	E-01
4	NIDS Detection Correlation	suricata.alert (Count: 41)	E-01, E-03
5	Zeek Connection State	SF (Normal SYN/FIN termination)	E-03

4.2 Dataset Event Breakdown

Event Dataset	Transport Protocol	Count	Analytical Description
zeek.conn	TCP	41	Total distinct, established TCP socket connection flows.
suricata.alert	TCP	41	Signature matches for Dridex JA3 hash (1 alert per connection flow).
zeek.ssl	TCP	41	Successful TLS cryptographic handshakes completed on port 2222.
zeek.notice	TCP	1	Zeek policy notice triggered during external communication.
Total Indexed Events:		124 events (41 distinct TCP connection sessions)	

4.3 Session Telemetry and Sample UIDs

Session UID	Source Socket	Destination Socket	Duration	State	Service
CQD5s1j5anUAsSGF2	10.11.28.101:50004	86.159.48.25:2222	2.14s	SF	ssl / tls
CQxeal1qTFU1DgPy1c	10.11.28.101:50009	86.159.48.25:2222	1.98s	SF	ssl / tls
Cfoe9tEq9LGC1zvr6	10.11.28.101:50110	86.159.48.25:2222	2.05s	SF	ssl / tls
CH1fzQ3KFWT1UnGU9	10.11.28.101:50124	86.159.48.25:2222	2.11s	SF	ssl / tls
CvghJ14KzEayP8ZNWg	10.11.28.101:50127	86.159.48.25:2222	2.08s	SF	ssl / tls

4.4 Command-and-Control Analysis

The 41 distinct TCP connections followed a highly structured heartbeat distribution. Rather than maintaining a single monolithic TCP socket, the malware engineered persistent control via sequential short-lived sessions connecting every ~180 seconds (3 minutes). This architectural pattern reduces network footprint anomalies associated with long-lived idle sessions while preserving interactive responsiveness.

4.5 Indicators of Compromise

Indicator	Type	Purpose / Assessment	Recommended Action
86.159.48.25	IPv4 (External)	Malicious C2 Endpoint (Port 2222)	Block all outbound traffic at firewall
10.11.28.101	IPv4 (Internal)	Infected Workstation	Isolate host from network; initiate remediation
TCP 2222	Network Port	Non-Standard C2 Port	Enforce strict egress filtering

4.6 Evidence Register

Evidence No.	Description	Source
E-01	Security Onion Hunt group metrics displaying exactly 41 <code>zeek.conn</code> and 41 <code>suricata.alert</code> events.	Security Onion Hunt (<code>01_tcp_2222_connection_count_41.png</code>)
E-02	CyberEx challenge portal confirmation validating flag <code>CTK{41}</code> conquered (997 pts).	CyberEx CTFd (<code>02challenged_solved.jpg</code>)
E-03	Security Onion Hunt events table showing individual Zeek connection and SSL records.	Security Onion Hunt (<code>03_zeek_conn_session_details.png</code>)

4.7 MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Evidence
Command and Control	Non-Standard Port	T1571	41 distinct outbound connections on TCP port 2222.
Command and Control	Encrypted Channel: Asymmetric Cryptography	T1573.002	TLSv1.2 encryption utilized across all 41 connection sessions.
Command and Control	Web Protocols: Periodic Beaconing	T1071.001	Automated ~3-minute connection polling cadence.

5. CONTAINMENT, ERADICATION AND RECOVERY

Immediate Actions

- 1. Isolate compromised host `10.11.28.101` to sever the 41-session recurring C2 loop with `86.159.48.25`.
- 2. Deploy border firewall rules dropping all outbound connections targeting `86.159.48.25` on all ports.
- 3. Audit Active Directory authentication logs during the time window of the 41 connections to identify any lateral movement attempts.

Additional Recommendations

- **Egress Filtering:** Implement a default-deny egress policy restricting internal endpoints from directly communicating over non-standard ports such as TCP 2222.
- **Host Re-imaging:** Perform a forensic disk/memory capture on `10.11.28.101` followed by a clean wipe and operating system reinstallation.
- **Signature Tuning:** Ensure JA3 hash fingerprint alerting (e.g., Dridex signatures) is configured to trigger automated blocking on perimeter network security appliances.

6. CONCLUSION

Forensic analysis of the Phase 4 PCAP dataset confirmed that compromised internal host `10.11.28.101` initiated exactly **41 distinct TCP connections** to external destination IP `86.159.48.25` on port `2222/TCP`. Each session completed a successful

TLS handshake, triggering 41 corresponding Suricata NIDS alerts for Dridex JA3 beaconing. Submission of flag `CTK{41}` was successfully verified and conquered on the CyberEx assessment portal.

7. EVIDENCE LIMITATIONS

- TLS encryption prevented inspection of inner HTTP/binary payloads within each of the 41 connection streams.
- Host-based endpoint telemetry (EDR/Sysmon) is required to determine the specific binary/process PID initiating the 41 outbound sockets.

8. APPENDICES

Appendix A - Investigation Queries

```
destination.ip:"86.159.48.25" AND destination.port:2222 | groupby event.dataset network.transport
```

```
destination.ip:"86.159.48.25" AND destination.port:2222 AND event.dataset:"zeek.conn"
```

Appendix B - Figures and Screenshots

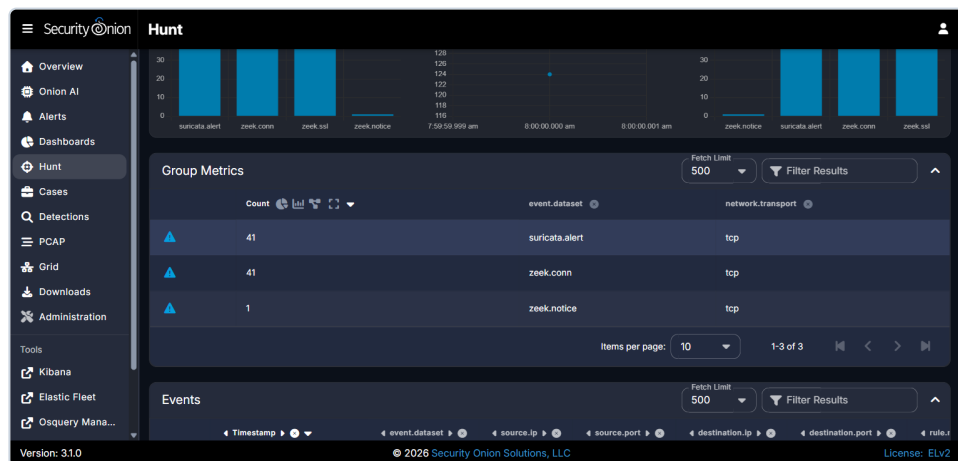


Figure 1: Security Onion Hunt aggregation query displaying exactly 41 zeek.conn and 41 suricata.alert events for destination 86.159.48.25:2222.

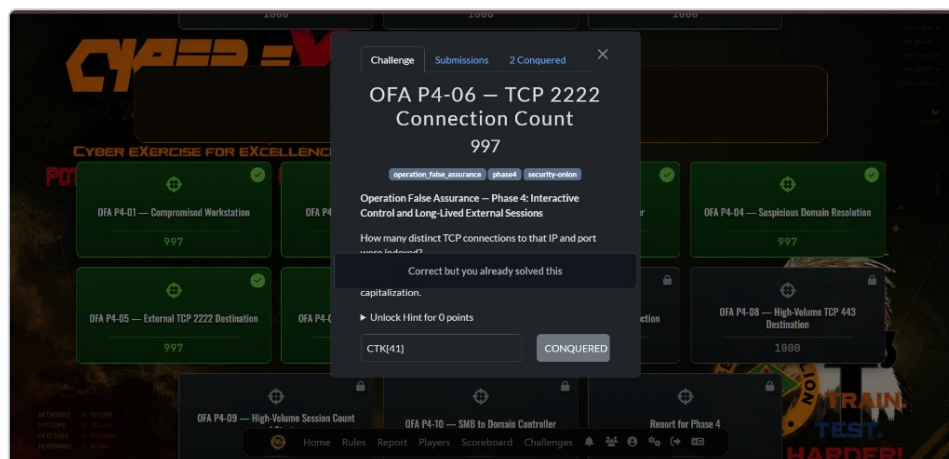


Figure 2: CyberEx challenge portal validation confirming OFA P4-06 TCP 2222 Connection Count solved with flag CTK{41} for 997 points.

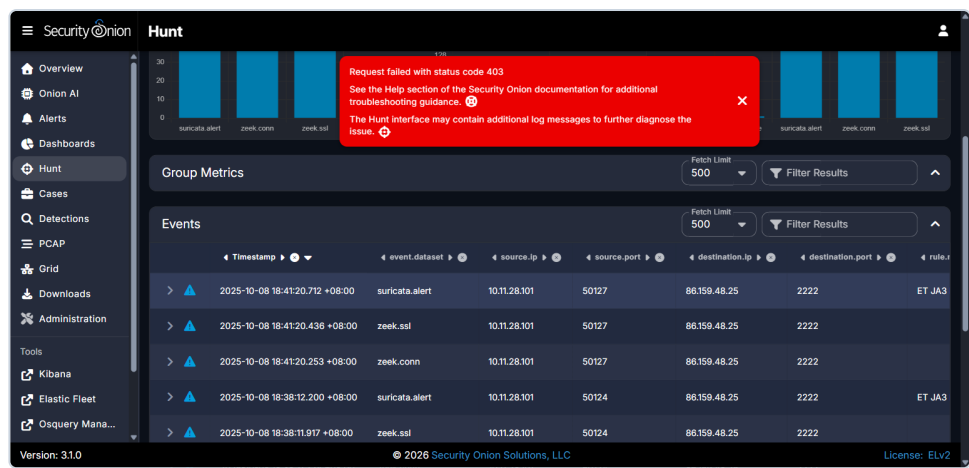


Figure 3: Security Onion Hunt event telemetry showing individual Zeek connection, SSL, and Suricata alert records.